

[Selenium: Fact Sheet for Consumers \(original version\)](#)

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Table of Contents

- [What is selenium and what does it do?](#)
- [How much selenium do I need?](#)
- [What foods provide selenium?](#)
- [What kinds of selenium dietary supplements are available?](#)
- [Am I getting enough selenium?](#)
- [What happens if I don't get enough selenium?](#)
- [What are some effects of selenium on health?](#)
- [Can selenium be harmful?](#)
- [Does selenium interact with medications or other dietary supplements?](#)
- [Selenium and healthful eating](#)
- [Where can I find out more about selenium?](#)
- [Disclaimer](#)
- [Glossary](#)

What is selenium and what does it do?

Selenium is a nutrient that the body needs to stay healthy. Your thyroid gland needs selenium to produce hormones and function properly. Selenium is also needed for reproduction, to make DNA, to help protect cells, and to resist infection.

How much selenium do I need?

The amount of selenium that you need each day depends on your age. Average daily recommended amounts are listed below in micrograms (mcg).

Life Stage	Recommended Amount
Birth to 6 months	15 mcg
Infants 7–12 months	20 mcg
Children 1–3 years	20 mcg
Children 4–8 years	30 mcg
Children 9–13 years	40 mcg
Teens 14–18 years	55 mcg
Adults	55 mcg
Pregnant teens and women	60 mcg
Breastfeeding teens and women	70 mcg

What foods provide selenium?

Selenium is found naturally in many foods, especially foods that contain protein. You can get the recommended amounts of selenium by eating a variety of foods, including:

- Fish and other seafood
- Beef, pork, poultry, eggs, and dairy products
- Breads, cereals, and other grain products
- Beans, lentils, and nuts

What kinds of selenium dietary supplements are available?

Selenium is found in multivitamin/mineral supplements and other dietary supplements. It can be present in several different forms, including selenomethionine, selenium-enriched yeast, sodium selenite, and sodium selenate. All dietary supplements include a Supplement Facts label, which will tell you how much selenium is in that product.

Am I getting enough selenium?

Most people in the United States get enough selenium from the foods they eat. However, certain groups of people may have trouble getting enough selenium:

- People undergoing kidney dialysis
- People living with HIV
- People who eat diets that consist mostly of plant foods grown in local soils that are low in selenium. This is because the amount of selenium in soil can affect the selenium content of foods.

What happens if I don't get enough selenium?

Selenium deficiency is very rare in the United States and Canada. Selenium deficiency can make iodine deficiency worse, increasing the risk of thyroid disease. Selenium deficiency can also cause Kashin-Beck disease (a type of arthritis that can begin in childhood) and Keshan disease (a type of heart disease that occurred mostly in the mid-20th century in China).

What are some effects of selenium on health?

Scientists are studying selenium to understand how it affects health. Here are some examples of what this research has shown.

Cancer

Getting enough selenium from your diet may lower your risk of some forms of cancer, including colorectal, prostate, lung, bladder, skin, esophageal, and gastric cancers.

Selenium supplements have been studied mostly in men to see if they reduce the risk of prostate cancer. These studies have found that selenium supplements do not lower the risk of prostate cancer, especially in men who get enough selenium from their diet.

More research is needed to understand the effects of selenium from food and dietary supplements on cancer risk.

Heart disease

Studies suggest that taking selenium supplements does not reduce the risk of heart disease, especially in people who get enough selenium from food. However, taking combination supplements that contain selenium and antioxidants, such as beta-carotene, vitamin C, or vitamin E, might lower the risk of dying of heart disease.

Cognitive decline and Alzheimer's disease

Some research has shown that people with low selenium levels might have higher risks of cognitive decline and Alzheimer's disease as they age, although most studies have not found that selenium supplements reduce these risks.

HIV infection

Children and adults with HIV who have selenium deficiency tend to have more severe disease. However, selenium supplements have shown little if any benefit for people with HIV.

While a few studies have found that pregnant women with HIV who take selenium supplements might have a lower risk of preterm birth, no other benefits have been found.

Male fertility

Selenium is an important nutrient for healthy sperm, and research shows that eating a healthy diet can improve sperm quality. However, it is unclear whether selenium supplements improve sperm quality.

More research is needed to determine whether selenium supplements affect male fertility.

Thyroid disease

Selenium is an important nutrient for the thyroid gland, and selenium levels are higher in the thyroid gland than in any other organ in the body. Research suggests that women who have low levels of selenium have a higher risk of thyroid disease, although a few studies have found that selenium supplements do not provide much benefit for the thyroid.

More research is needed to understand the effects of selenium on thyroid disease.

Can selenium be harmful?

Yes, if you get too much selenium over time. Brazil nuts, for example, contain very high amounts of selenium (68–91 mcg per nut) and can cause you to go over the upper limit if you eat too many. Getting excessive amounts of selenium from dietary supplements can also be harmful.

Getting too much selenium can cause:

- Garlic breath
- Hair loss
- Skin rash
- Diarrhea
- Irritability
- Metallic taste in the mouth
- Brittle nails or nail loss
- Nausea
- Extreme tiredness
- Nervous system problems

Consuming very high amounts of selenium can cause severe stomach and nervous system problems, difficulty breathing, tremors, lightheadedness, flushing, muscle tenderness, kidney or heart failure, and heart attacks. In rare cases, it can cause death.

The daily upper limits for selenium include intakes from all sources—food, beverages, and supplements—and are listed below. These upper limits do not apply if you are taking selenium under the care of a doctor.

Ages	Upper Limit
Birth to 6 months	45 mcg
Infants 7–12 months	60 mcg
Children 1–3 years	90 mcg
Children 4–8 years	150 mcg
Children 9–13 years	280 mcg
Teens 14–18 years	400 mcg
Adults	400 mcg
Pregnant teens and women	400 mcg
Breastfeeding teens and women	400 mcg

Some experts recommend daily upper limits for selenium that are lower than these amounts. For example, in 2023 the European Food Safety Authority set a daily upper limit for selenium of 255 mcg for all adults, with lower amounts ranging from 70 to 230 mcg for children and teens, depending on age.

Does selenium interact with medications or other dietary supplements?

Yes, selenium may interact with certain medications, and some medications may interact with selenium. For example, cisplatin, a chemotherapy drug that is used to treat cancer, can lower selenium levels, but the effect this has on the body is not clear.

Tell your doctor, pharmacist, and other health care providers about any dietary supplements and prescription or over-the-counter medicines you take. They can tell you if the dietary supplements might interact with your medicines or if the medicines might interfere with how your body absorbs, uses, or breaks down nutrients.

Selenium and healthful eating

People should get most of their nutrients from food and beverages, according to the federal government's Dietary Guidelines for Americans. Foods contain vitamins, minerals, dietary fiber, and other components that benefit health. In some cases, fortified foods and dietary supplements are useful when it is not possible to meet needs for one or more nutrients (for example, during specific life stages such as pregnancy). For more information about building a healthy dietary pattern, see the [Dietary Guidelines for Americans](#).

Where can I find out more about selenium?

- For general information on selenium
 - [Health Professional Fact Sheet on Selenium](#), Office of Dietary Supplements (ODS)
 - [Selenium in Diet](#), MedlinePlus
- For more information on food sources of selenium
 - [FoodData Central](#), USDA
 - Nutrient list for selenium (listed by [food](#) or by [selenium content](#)), USDA
- For more advice on choosing dietary supplements
 - [Frequently Asked Questions: Which brand\(s\) of dietary supplements should I purchase?](#), ODS
- For information about building a healthy dietary pattern
 - [Dietary Guidelines for Americans](#)

Glossary

absorption

In nutrition, the process of moving protein, carbohydrates, fats, and other nutrients from the digestive system into the bloodstream. Most absorption occurs in the small intestine.

Alzheimer's disease

A brain disease in which thinking, memory, and reasoning ability is slowly destroyed. In advanced stages, an affected person becomes disoriented and confused, has mood and behavior changes, and has difficulty talking, walking, and swallowing. Alzheimer's disease is progressive, irreversible, and incurable.

antioxidant

A substance that protects cells from damage caused by free radicals (compounds formed during the metabolism of oxygen). It may help prevent the development of some chronic diseases such as cancer. Antioxidants include beta-carotene; lutein; lycopene; vitamins A, C, and E; selenium; and zinc.

arthritis

A group of diseases in which one or more joints (places in the body where two bones connect) become swollen and painful. The most common type of arthritis is osteoarthritis. It is caused by the breakdown of cartilage, a type of tissue that cushions and supports the joint. Without cartilage, the bones in the joint rub together, causing inflammation (swelling, redness, pain, and warmth) and stiffness. Arthritis may affect the fingers, hips, knees, lower back, feet, or any joint in the body.

beta-carotene

A carotenoid found in carrots, cantaloupe, apricots, sweet potatoes, pumpkin, winter squash, mangos, collard greens, spinach, kale, broccoli, and other orange, red, and dark green fruits and vegetables.

cancer

A group of diseases in which cells divide abnormally and without control, and spread to nearby tissues and other parts of the body. Without treatment, cancer can stop organs from working normally, damage body systems, and cause the patient to die. Cancer may be caused by multiple factors, such as radiation, sunlight, tobacco, certain viruses, and poisonous chemicals; however, the cause of many cancers is unknown.

cell

The individual unit that makes up the tissues of the body. All living things are made up of one or more cells, which are the smallest units of living structure capable of independent existence.

chemotherapy

A chemical that kills bacteria, viruses, fungi, or tumor cells. It usually refers to drugs used in cancer treatment.

dairy food

Milk and products made with milk, such as buttermilk, yogurt, cheese, cottage cheese, and ice cream.

deficiency

An amount that is not enough; a shortage.

deoxyribonucleic acid

DNA. The molecules inside cells that carry genetic information and pass it from one generation to the next.

dialysis

The process of filtering the blood when the kidneys are not able to cleanse it.

diarrhea

Loose, watery stools.

dietary fiber

A substance in plants that you cannot digest. It adds bulk to your diet to make you feel full, helps prevent constipation, and may help lower the risk of heart disease and diabetes. Good sources of dietary fiber include whole grains (such as brown rice, oats, quinoa, bulgur, and popcorn), legumes (such as black beans, garbanzo beans, split peas, and lentils), nuts, seeds, fruit, and vegetables.

Dietary Guidelines for Americans

Advice from the federal government to promote health and reduce the chance (risk) of long-lasting (chronic) diseases through nutrition and physical activity. The Guidelines are updated and published every 5 years by the US Department of Health and Human Services and the US Department of Agriculture.

dietary supplement

A product that is intended to supplement the diet. A dietary supplement contains one or more dietary ingredients (including vitamins, minerals, herbs or other botanicals, amino acids, and other substances) or their components; is intended to be taken by mouth as a pill, capsule, tablet, or liquid; and is identified on the front label of the product as being a dietary supplement.

drug

Any substance (other than food) that is used to prevent, diagnose, treat, or relieve symptoms of a disease or abnormal condition. Also, a substance that alters mood or body function or that can be habit-forming or addictive, especially a narcotic.

enriched

When certain nutrients (such as vitamins and minerals) have been added to a food product to replace nutrients that may be lost during processing or storage. For example, white flour is enriched with thiamin, riboflavin, niacin, and iron because those nutrients are lost when grain is made into flour. Folic acid is also added to enriched flour.

esophagus

The muscular tube through which food passes from the throat to the stomach.

fortified

When nutrients (such as vitamins and minerals) are added to a food product. For example, when calcium is added to orange juice, the orange juice is said to be "fortified with calcium". Similarly, many breakfast cereals are "fortified" with several vitamins and minerals.

gastric cancer

Uncontrolled growth of abnormal cells in the stomach that spreads through the outer layers of the stomach as it grows.

gland

A small organ that makes and releases a substance such as sweat, tears, saliva, milk, a hormone, or substances that aid in digestion.

health care provider

A person who supplies health care services. Health care providers include individuals with professional training (including doctors, nurses, technicians, and aides).

heart attack

The blockage of an artery supplying blood and oxygen to the heart, resulting in the damage or death of a section of heart muscle.

heart failure

A condition in which the heart is unable to pump the amount of blood needed by the body. It is caused by high blood pressure, heart attack, and other disorders of the heart or blood vessels. Also called congestive heart failure.

hormone

A group of chemicals made by glands in the body. Hormones circulate in the bloodstream and control the actions of certain cells or organs. Some hormones can also be manufactured.

infant

A child younger than 12 months old.

infection

The invasion and spread of germs in the body. The germs may be bacteria, viruses, yeast, or fungi.

interaction

A change in the way a dietary supplement acts in the body when taken with certain other supplements, medicines, or foods, or when taken with certain medical conditions. Interactions may cause the dietary supplement to be more or less effective, or cause effects on the body that are not expected.

iodine

A mineral the body needs to make thyroid hormones, which control metabolism (the process of turning the food you eat into energy your body can use) and many other essential functions, including bone and brain development during pregnancy and infancy. Iodine is found in seaweed, seafood, dairy products, grain products, eggs, and iodized salt.

kidney

One of two organs that remove waste from the blood (as urine). The kidneys also make erythropoietin (a substance that stimulates red blood cell production) and help regulate blood pressure. The kidneys are located near the back under the lower ribs.

label

When referring to dietary supplements, information that appears on the product container, including a descriptive name of the product stating that it is a "supplement"; the name and place of business of the manufacturer, packer, or distributor; a complete list of ingredients; and each dietary ingredient contained in the product. Supplements must also include directions for use, nutrition labeling in the form of a Supplement Facts panel that identifies each dietary ingredient contained in the product and the serving size, amount, and active ingredients.

lung

An organ in the chest that supplies oxygen to the body and removes carbon dioxide. There are two lungs in the body.

microgram

μg or mcg. A unit of weight in the metric system equal to one millionth of a gram. (A gram is approximately one-thirtieth of an ounce.)

mineral

In nutrition, an inorganic substance found in the earth that is required to maintain health.

multivitamin/mineral dietary supplement

MVM. A product that is meant to supplement the diet. MVMs contain a variety of vitamins and minerals. The number and amounts of these nutrients can vary substantially by product.

nausea

The uneasy feeling of having an urge to throw up (vomit).

nervous system

The brain and spinal cord, including the network of nerves that carry messages back and forth between the brain and all parts of the body. The nervous system controls what the body does.

nutrient

A chemical compound in food that is used by the body to function and maintain health. Examples of nutrients include proteins, fats, carbohydrates, vitamins, and minerals.

Office of Dietary Supplements

ODS, Office of Disease Prevention, Office of Director, National Institutes of Health, Department of Health and Human Services. ODS strengthens knowledge and understanding of dietary supplements by evaluating scientific information, stimulating and supporting research, disseminating research results, and educating the public to foster an enhanced quality of life and health for the US population.

pharmacist

A person licensed to make and dispense (give out) prescription drugs and who has been taught how they work, how to use them, and their side effects.

poultry

Birds that are raised for eggs or meat, including chickens, turkeys, ducks, and geese.

prescription

A written order from a health care provider for medicine, therapy, or tests.

prostate cancer

Uncontrolled growth of abnormal cells in the prostate (a gland in the male reproductive system found below the bladder and in front of the rectum). Prostate cancer usually occurs in older men.

protein

A molecule made up of amino acids that the body needs for good health. Proteins are the basis of body structures such as skin and muscle, and substances such as enzymes and antibodies.

risk

The chance or probability that a harmful event will occur. In health, for example, the chance that someone will develop a disease or condition.

selenium

A mineral required in very small amounts to make important enzymes that are essential for good health. Selenium is found in some foods, including plant foods grown in selenium-rich soil, and some meats and seafood.

supplement

A nutrient that may be added to the diet to increase the intake of that nutrient. Sometimes used to mean dietary supplement.

thyroid

A gland located in the front of the neck, below the larynx (Adam's apple). The thyroid makes hormones that circulate in the bloodstream and affect brain development, metabolism, weight, breathing, heart rate, blood pressure, nervous system functions, body temperature, muscle strength, skin dryness, menstrual cycles, and cholesterol levels.

treat

To care for a patient with a disease by using medicine, surgery, or other approaches.

tremor

A trembling or shaking in one or more parts of the body, usually the hands. An individual can also have tremors in the arms, head, face, vocal cords, torso, and legs.

upper limit

UL. The largest daily intake of a nutrient considered safe for most people. Taking more than the UL is not recommended and may be harmful. The UL for each nutrient is set by the Food and Nutrition Board at the National Academies of Sciences, Engineering, and Medicine. For example, the UL for vitamin A is 3,000 micrograms/day. Women who consume more than this amount every day shortly before or during pregnancy have an increased chance (risk) of having a baby with a birth defect. Also called the tolerable upper intake level.

US Department of Agriculture

USDA promotes America's health through food and nutrition, and advances the science of nutrition by monitoring food and nutrient consumption and updating nutrient requirements and food composition data. USDA is responsible for food safety, improving nutrition and health by providing food assistance and nutrition education, expanding markets for agricultural products, managing and protecting US

public and private lands, and providing financial programs to improve the economy and quality of rural American life.

vitamin

A nutrient that the body needs in small amounts to function and maintain health. Examples are vitamins A, C, and E.

vitamin C

A nutrient needed by the body to make collagen (a protein found in cartilage, tendons, ligaments, bone, and blood vessels), to absorb iron from food, and for wound healing. It is an antioxidant and protects cells from free radical damage. Vitamin C is found in some foods including citrus fruits, strawberries, peppers, dark green vegetables, tomatoes, and potatoes. Also called ascorbic acid.

vitamin E

A nutrient needed by the body to help keep the immune system healthy and to repair damage to DNA. It is an antioxidant that protects cells from free radical damage. Vitamin E is found in some foods, including vegetable oils, nuts and seeds, fortified breakfast cereals, and spinach, broccoli, kiwi, and mangos.

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